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Estimate Logit Model and Calculate Marginal Effects



Estimate and view logit model. Here X is the warranty discount. Y=1 if the warranty is purchased and 0 otherwise.

```
# Estimate logit model for AWE data.
AME_logit = tpstats.model("logit", "Y~X", AWE_warranty)
print(AME_logit.summary())
```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-3.4280	0.268	-12.806	0.000	-3.953	-2.903
X	0.0532	0.007	7.679	0.000	0.040	0.067

```
# Calculate average marginal effect
AME_logit.marginal_effects()
```

We obtain .0066 as the marginal effect.

Hence, a 1 percentage point increase in the warranty discount increases the predicted purchase probability by .66 percentage points.

This is the average effect. The exact effect varies with the value of X.

[Optional Derivation of Likelihood Function for Logit Model](#)

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Calculating Probability from Logit Equation



What is the predicted probability that a customer who buys a refrigerator will purchase the warranty if we offer a discount of 45%?

Recall that the logit probability is:

$$\pi_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

Plugging in our coefficients from the previous page: $\hat{\pi}_i = \frac{e^{-3.428 + 0.0532x_i}}{1 + e^{-3.428 + 0.0532x_i}}$

Inserting our discount of 45, we find the probability of purchase is: .26 (26%).

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Predicted Effects of 1 Percentage Point Change



To find the effect of a change of X on π , you can let the computer do the work.

Recall that X denotes the warranty discount.

For example, to find the effect of increasing X from 10 to 20:

```
# Change in probability of purchase of increasing discount from 10% to 20%
Xnew_10_20 = pandas.DataFrame({"X": [10, 20]})
AME_logit.predict(Xnew_10_20)
```

We obtain:

0	0.052371
1	0.086022

- The purchase probabilities for discounts 10% and 20% are .0524 and .0860.
- Hence, increasing the discount from 10% to 20% increases the predicted purchase probability, $\hat{\pi}_i$, by $(.0860 - .0524) = .0336$ (i.e., 3.36 percentage points).
- Note the magnitude of the purchase probability depends on the point on the curve.
- To enhance your understanding, investigate a change in discount from 25% to 35% and report the increase in purchase probability in percentage points.

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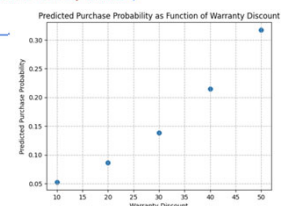
Predicted Probabilities



Calculate predicted purchase probabilities for the values of X used in our experiment and plot predicted probabilities as function of warranty discount, X:

```
# Calculate and plot predicted probabilities from experiment
AME_Pred = AME_logit.predict()
plt.scatter(AME_warranty["X"], AME_Pred)
plt.grid(True, linestyle='--')
plt.title("Predicted Purchase Probability as Function of Warranty Discount")
plt.xlabel("Warranty Discount")
plt.ylabel("Predicted Purchase Probability")
```

We see that the predicted purchase probability increases to 32 percent with a discount of 50%.



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Classification Table

Our classification table is shown below.

- The default `p_cutoff` is the fraction of ones in our data, `.159`.
- With `p_cutoff=.159`, the model correctly predicts 0 for 550 observations and correctly predicts 1 for 97 observations for a total of 647 correct predictions.
- The model incorrectly predicts 0 for 62 observations and incorrectly predicts 1 for 291 observations for a total of 353 incorrect predictions.

```

AWE_logit.classification_table()
# Correct predictions: 550+97, incorrect prediction 291+62
[550.0, 291.0, 62.0, 97.0]

```

	Summary	Actual	Predicted	Frequency
0	Correct	0	0	550.0
1	Incorrect	0	1	291.0
2	Incorrect	1	0	62.0
3	Correct	1	1	97.0

Note that AWE has no financial outlay for incorrect predictions because AWE is only liable for the cost of servicing a warranty if a customer buys the warranty.

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Compare Linear Probability Model to Logit

Estimate a linear regression (linear probability model):

```

AWE_LPM=lmprstats.model("cs", "Y ~ X", AWE_warranty)
AWE_LPM.summary()

```

OLS Regression Results

Dep. Variable:	Y	R-squared:	0.065
Model:	OLS	Adj. R-squared:	0.064
No. Observations:	1000	F-statistic:	66.14
Covariance Type:	HC1	Prob (F-statistic):	1.24e-15

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-0.0358	0.021	-1.698	0.090	-0.077	0.006
X	0.0066	0.001	8.133	0.000	0.005	0.008

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Note the ease of interpretation of the coefficient of X:

- A one unit increase in X increases the predicted probability of purchase by .006 (i.e., .66 percentage points).

Predicted probabilities of logit and linear probability model are compared on next slide.

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Comparing Logit and Regression Predictions

- The strength of the linear probability model (i.e., linear regression) for a (0,1) dependent variable is ease of interpretation.
- The weakness of the linear probability model is illustrated in this graph.
- The linear probability model over-predicts for intermediate discounts and underpredicts for high and low discounts.
- In some applications, the linear probability model performs well (e.g. the Airbnb case).
- The linear probability model is clearly unsuitable for this application.

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Analysis of The AWE Decision

- AWE has been pricing its warranty at \$300.
- If it adopts a discount of $x\%$, its price will be $300*(1-x/100)$.
- Hence, its revenue from each warranty sold will be $300*(1-x/100)$.
- From past experience, AWE knows that the present value of the cost of servicing a warranty over five years will be \$120.
- Hence, with a discount of $x\%$, AWE will earn $(300*(1-x/100)-120)$ of profit per warranty.
- Let $p(x)$ be the probability that a customer will purchase the warranty for a discount of x .
- Then the expected profit per customer with discount x will be:

$$(300*(1-x/100)-120)*p(x).$$

We will now see that this is easily done in Python.

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Finding Optimal Warranty Price

We will evaluate profit at several possible values of the warranty discount. Steps in calculation follow.

```
# Create discount percent values from 0 to 50, name Discpct and put in a dataframe named Discpct
Discpct=numpy.arange(51)
Discpct = pandas.DataFrame({"X": Discpct})

# Calculate profit per purchase for each value of discount
Profit_Per_Purchase=(300 * (1 - Discpct/100) - 120)

# Predict purchase probabilities for each value of discount using logit model
Purchase_Probs=logit.predict(Discpct)
# Calculate expected profit for each discount
Profit=Profit_Per_Purchase.*Purchase_Probs

# Put Purchase_Probs in a dataframe
Purchase_Probs = pandas.DataFrame({"X": Purchase_Probs})

# Calculate warranty price for each discount
Price=300 * (1 - Discpct/ 100)

# Calculate predicted profit
Profit=(Price-120)*Purchase_Probs
```

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Choosing The Optimal Discount



We see the optimal discount is approximately 35%.
To find the precise amount, proceed as follows.

```
# Put key variables in a data frame
Key_Variables=pandas.concat([Discpct,Purchase_Probs, Profit,Price], axis=1)
# Name the columns in Key_Variables
Key_Variables.columns=['Discpct','Purchase_Probs','Profit','Price']
# Round all values to 3 digits after decimal point
Key_Variables=Key_Variables.round(3)
# View the results. Scroll down to find discount that maximizes profit
print(Key_Variables)
```

Now scroll down in the output produced by the above print command.

You will see that the result is a discount of 37% and you will see the expected profit at this discount is \$13.02 per refrigerator sold.

The optimal warranty price is then $300 \times (1 - 37/100) = \189 .

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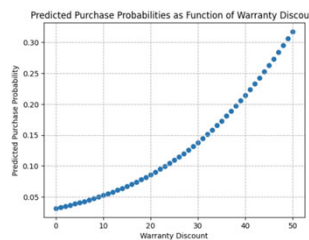
Building Intuition

Why does the curve on the previous slide first increase and then turn downward?

To help you think about this, here is a plot of predicted purchase probabilities as a function of discount.

We see that purchase probability increases as the discount increases.

What happens to profit per warranty as the warranty discount increases?



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Summary

This application demonstrates the value of combining the following for a managerial decision:

- Conducting an Experiment
- Logit Estimation
- Optimization

Note that the type of experiment utilized in this application can be conducted as part of normal business operations.

Hence, the approach presented in this case can be applied in a variety of managerial settings.

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This appendix builds on our derivation in the appendix to the solar sales application. You might want to review that appendix before working through this appendix.

Let $y_i = 1$ if the individual i purchased the warranty and $y_i = 0$ if the individual declined the warranty.

Let x_i be the percent discount offered to customer i .

Let's write the probability that individual i buys the warranty as follows:

$$1. \quad \pi_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

Now, a typical element of our likelihood function is:

$$2. \quad f(y_i) = \pi_i^{y_i} (1 - \pi_i)^{(1-y_i)} = \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{y_i} \left(1 - \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{(1-y_i)}$$

Hence, our logit likelihood function is the product of n terms like (9):

$$3. \quad L(y_1, y_2, \dots, y_n) = f(y_1) f(y_2) \dots f(y_n) \\ = \left(\frac{e^{\beta_0 + \beta_1 x_1}}{1 + e^{\beta_0 + \beta_1 x_1}} \right)^{y_1} \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_1}} \right)^{(1-y_1)} \dots \left(\frac{e^{\beta_0 + \beta_1 x_n}}{1 + e^{\beta_0 + \beta_1 x_n}} \right)^{y_n} \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_n}} \right)^{(1-y_n)}$$

We substitute in the y_i and x_i .

The Maximum Likelihood Estimations are then the values of β_0 and β_1 that maximize (3).

We do let the computer do the work.

When you issue the logit command in R, it choose the estimates of β_0 and β_1 to maximize (3) and reports the results.